MILITARY SPECIFICATION

PROJECTILE, 20MM, API-T, M52E1, METAL PARTS

1. SCOPE

1.1 This specification covers the metal parts and body assembly for Projectile, 20MM, API-T, M52E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-651 - Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

DRAWINGS

MUNITIONS COMMAND

C7258808	- Adapter
C7258809	- Nose
B7258810	- Disc, Closure
B7258816	- Blank, Rotating Band, for Projectiles, 20MM
F7258861	- Carton for Interplant Shipment of Projectiles, 20MM, Assemblies and Metal Parts Assemblies (200)
D7259220	- Projectile, 20MM, API-T, M52E1
D7259221	- Body, Adapter and Rotating Band Assembly
D7259222	- Body
D7259223	- Disc, Metering
D7259136	- Shipping Container for Inert Projectile, 20MM, Assemblies and Metal Parts Assemblies (10,000)
D7259192	- Cartridge, 20MM, for Projectile, API or API-T, Metal Parts Security Test
IEL 7259220	- Index of Inspection Equipment Lists for: Projectile, 20MM, API-T, M52E1, Metal Parts

PUBLICATIONS

MILITARY

AMCR 715-505 -- Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-All parts and assemblies shall comply with the requirements of applicable drawings and specifications.

3.2 First article testing.-Requirements for submission of the first article sample by the contractor shall be as specified in the contract. (see 6.2)

3.3 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.3)

3.4 Hardness.-Hardness of the body, the nose, and the adapter shall comply with the respective requirements of Dwgs. D7259222, C7258809, and D7259221.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

3.5 Adapter security.-The adapter shall remain secured to the body when subjected to the tension loading specified on Dwg. D7259221.

3.6 Banding.-The rotating band and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used for cleaning the band seat after knurling. The rotating band blank shall fill the band seat. The average radial void between the seated band and the band seat knurling shall not exceed 0.004 inch.

3.7 Soundness.-The body, adapter and rotating band assembly shall be free of ferrous metal discontinuity detectable by magnetic particle inspection.

3.8 Protective coating.

3.8.1 Phosphate coating.-The pretreatment process used shall comply with applicable requirements of Dwg. D7259220. The phosphate coating deposit shall comply with the requirements of TT-C-490 for appearance and minimum weight as applicable to the type of phosphate coating.

3.8.2 Paint coating.-The paint coatings shall comply with applicable requirement of Dwg. D7259220. Exterior paint adhesion shall comply with applicable requirements of TT-C-490.

3.8.3 Corrosion resistance.-The painted assembly shall show no visible evidence of ferrous metal corrosion after exposure to 5% salt spray solution for 24 hours.

3.9 Metal parts security.-The body, adapter and rotating band assembly, loaded and assembled in accordance with Dwg. D7259192, shall not separate or breakup in the test for metal parts security.

3.10 Workmanship.-The requirements for workmanship are as specified in applicable drawings, referenced specifications, and the following:

3.10.1 Processing defects.-The parts and assemblies shall be free of cracks, cuts, dents, burrs, sharp edges, excessive surface roughness and other forms of substandard workmanship.

3.10.2 Cleanliness.-The parts and assemblies shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Manufacturing process.-Submission of the contractor's proposed manufacturing process and subsequent changes for approval shall be as directed by the contracting officer.

4.3 First article sample.

4.3.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

a. Inspection and nonfiring tests.

(1) 100 body-adapter and rotating band assemblies (painted)
(2) 100 body-adapter and rotating band assemblies (unpainted)
(3) 100 unbanded bodies
(4) 100 adapters
(5) 25 rotating band blanks
(6) 100 closure discs
(7) 100 noses
(8) 100 metering discs (formed)
(9) 100 metering discs (before forming)

b. Firing test.

(1) 4000 body-adapter and rotating band assemblies (painted)
(2) 4000 noses
(3) 4000 closure discs
(4) 4000 metering discs (formed)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production.

4.3.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract. (see 6.2)

4.3.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.4 Inspection provisions.

4.4.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.4.1.1 Lot.-A lot shall consist of projectile body assemblies (painted), produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision. (see 3.3)

4.4.1.2 Lot identification.-Each projectile body assembly (painted), and each packed lot of metal parts shall be identified in accordance with applicable drawings and MIL-STD-1168, supplemented as directed by the procuring agency.

4.4.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each part or assembly found to be defective shall be rejected.

4.4.2.1 Sampling and acceptance criteria.-Sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235, except as otherwise indicated. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the AQL assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, when three or less defect characteristics are listed in a classification or, when sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects. (see 6.5)

4.4.2.2 AQL's.-The following AQL's are assigned:

a. Individual defect characteristic basis.

Major defects	0.25%
Minor defects	0.40%

b. Defect characteristic, class basis.

Major defects	1.50%
Minor defects	2.50%

c. Individual defect characteristic associated with the class basis.

Major defects	0.40%

4.4.3 Classification of defects 1/.-The classification of defects shall be as follows: (see 6.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.4.3.1 Metering disc before forming (see Dwg. C7259223).

Categories and defects	Method of inspection
Critical: None defined.	
Major:	
101. Outside diameter (O.D.)	Gage
102. Thickness	Gage
Minor:	
201. Inside diameter	Gage
202. O.D. not perpendicular to face of disc	Gage

4.4.3.2 Metering disc, formed (see Dwg. C7259223).

Categories and defects	Method of inspection
Critical: None defined.	
Major:	
101. O.D.	Gage
Minor:	
201. Hole missing	Visual
202. Tear, crack or split extending thru thickness of metal	Visual
203. Protective finish missing or incomplete	Visual
204. Foreign matter	Visual

4.4.3.3 Rotating band blank (see Dwg. B7258816).

Categories and defects	Method of inspection
Critical: None defined.	
Major:	
101. Foreign matter	Visual (see 3.6)
102. Defective metal, including laminations	Visual
103. Incomplete or deformed	Visual 2/
Minor:	
201. Width	Gage

4.4.3.4 Body (see Dwg. D7259222).

Categories and defects	Method of inspection
Critical: None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

Major:

101.	Rotating band seat knurling missing or incomplete	Visual
102.	Adapter seat grooving or knurling missing or incomplete	Visual/Gage 3/
103.	Diameter adapter seat after knurling, min.	Gage
104.	Length, base to datum diameter	Gage
105.	Length, datum diameter to adapter seat shoulder	Gage
106.	Concentricity, adapter seat with base O.D.	Gage
107.	Depth, rotating band seat, knurling, min.	Visual/Gage 4/
108.	Diameter, tracer cavity, min.	Gage
109.	Concentricity, tracer cavity with base O.D.	Gage
110.	Diameter, metering disc cavity	Gage
111.	Concentricity, metering disc cavity with base O.D.	Gage
112.	Width, metering disc cavity	Gage
113.	Location, metering disc cavity	Gage
114.	Diameter of retaining flange, metering disc	Gage
115.	Concentricity, with base O.D. of retaining flange, metering disc	Gage

Minor:

201.	Length, datum diameter to rear of rotating band seat	Gage
202.	Diameter, rotating band seat, knurled	Gage
203.	Width, rotating band seat	Gage
204.	Weight	Balance
205.	Flat at nose, max.	Visual
206.	Finish, ogive and tracer cavity	Visual/Feel
207.	Width, rotating band seat knurling, min.	Visual/Gage 4/
208.	Foreign matter	Visual (see 3.6)

4.4.3.5 Adapter (see Dwg. C7258808).

Categories and defects	Method of inspection

Critical: None defined.

Major:

101.	Large inside diameter (I.D.) max.	Gage
102.	Concentricity, I.D. of serrations with large I.D.	Gage
103.	Concentricity, large I.D. with outside diameter (O.D.)	Gage
104.	Depth, large I.D.	Gage

Minor:

201.	Finish	Visual/Feel
202.	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.4.3.6 Disc, closure (see Dwg. B7258810).

Categories and defects	Method of inspection
Critical: None defined.	
Major:	
101. External profile, except overall length	Gage
102. Length to datum diameter	Gage
103. Concentricity, datum diameter with datum surface	Gage
104. Concentricity, large O.D. with datum surface	Gage
105. Depth, datum diameter to cavity bottom, min.	Gage
Minor:	
201. Overall length	Gage
202. Finish	Visual/Feel
203. Foreign matter	Visual

4.4.3.7 Nose (see Dwg. C7258809).

Categories and defects	Method of inspection
Critical: None defined.	
Major:	
101. Metal defective	Visual (see 3.10)
102. Inside diameter, straight section	Gage
103. Wall thickness, closed end	Gage
104. Foreign matter on interior	Visual
105. Concentricity, large diameter base end with tapered surface	Gage
106. Profile and length, shoulder to open end	Gage
107. Profile and length, shoulder to closed end	Gage
Minor:	
201. Finish	Visual/Feel

4.4.3.8 Body, adapter and rotating band assembly (see Dwg. D7259221).

Categories and defects	Method of inspection
Critical:	
1. Ferrous metal discontinuity	Soundness test (see 4.5.4)
Major:	
101. Overall length, min.	Gage
102. Depth, to body datum diameter	Gage
103. Diameter of bourrelet, min.	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

104.	Concentricity, body rear of rotating band with bourrelet	Gage
105.	Concentricity, ogive with bourrelet	Gage
106.	Concentricity, rotating band with bourrelet	Gage
107.	Length, rear of rotating band to open end of adapter	Gage
108.	Profile, crimp groove	Gage
109.	Radius, rear edge of crimp groove	Gage
110.	Diameter, rotating band	Gage
111.	Serrations in adapter missing or incomplete	Visual
112.	Location, crimp groove	Gage
113.	Width, rotating band	Gage
114.	Inside diameter across serrations	Gage
115.	Weight	Balance
116.	Diameter, body, rear of rotating band, min.	Gage
117.	Unfilled band seat (gap)	Visual

Minor:

201.	Finish	Visual/Feel
202.	Mouth counterbore missing	Visual
203.	Depth and width of undercut, front and rear of rotating band	Gage
204.	Base radius missing	Visual
205.	Overall length, max.	Gage
206.	O.D., adapter end	Gage
207.	Corner break at adapter end, max.	Visual

4.4.3.9 Body, adapter and rotating band assembly (painted)(see Dwg. D7259220).

Categories and defects		Method of inspection

Critical: None defined.

Major:

101.	Diameter, bourrelet, max.	Gage
102.	Diameter, body, rear of rotating band, max.	Gage
103.	Rotating band damaged	Visual
104.	Diameter, tracer cavity, min.	Gage
105.	Depth, tracer cavity, min.	Gage

Minor:

201.	Paint not dry	Tactile
202.	Marking incorrect, incomplete, illegible or missing--	Visual
203.	Paint on rotating band	Visual
204.	Paint inadequate, indicated exterior surface and base cavity	Visual/Feel
205.	Foreign matter	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

2/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions, shall be classed defective.

3/ Measurement of grooving for compliance with requirements for location and depth shall apply where visual inspection indicates that the location or depth dimension is borderline.

4/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

4.4.4 Tests.-The tests listed in Table I shall be conducted on each lot in accordance with the test methods and procedures of 4.5. For acceptance, the results of each test shall comply with the applicable requirement(s).

4.4.4.1 Test samples.-The sample sizes and acceptance criteria for the various tests shall be as specified in Table I.

Table I

Test	Sample Size		Item	Req. Ref.
Hardness				
Body	1/	15	Body	3.4
Adapter	2/	75	Body & adapter assembly, unbanded	3.4
Nose	3/	75	Nose	3.4
Adapter security	4/	20	Body & adapter assembly, unbanded	3.5
Band tightness	5/	300	Body, adapter & rotating band assembly (unpainted)	3.6
Soundness verification	6/	300	Body, adapter & rotating band assembly (unpainted)	4.5.5
Magnetic particle inspection effectiveness	7/	20,Min	Inspection effectiveness standard, unscheduled frequency	4.5.6
Phosphate coating	8/	3	Standard panels, TT-C-490	3.8.1
Paint adhesion	9/	15	Body, adapter & rotating band assembly (painted)	3.8.2
Corrosion resistance	10/	30	Body, adapter & rotating band assembly (painted)	3.8.3
Metal parts security	11/	250	Body, adapter & rotating band assembly (painted)	3.9

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

1/ If three or more sample bodies fail to meet the hardness requirements of Dwg. D7259222, the lot shall be rejected. If one or two bodies fail the hardness requirements, a second sample consisting of double the number of bodies in the first shall be tested. If in the combined samples three or more bodies fail, the lot shall be rejected.

2/ If three or more sample adapters fail to meet the hardness requirement of Dwg. D7259221, the lot shall be rejected. If one or two adapters fail the hardness requirements, a second sample consisting of double the number of adapters in the first shall be tested. If in the combined samples three or more adapters fail, the lot shall be rejected.

3/ If three or more sample noses fail to meet the hardness requirement of Dwg. D7258809, the lot shall be rejected. If one or two noses fail the hardness requirement, a second sample consisting of double the number of noses in the first shall be tested. If in the combined samples three or more noses fail, the lot shall be rejected.

4/ If one or more sample adapters fail to the adapter security requirement, the lot shall be rejected.

5/ If two or more assemblies of the sample indicate failure to comply with the applicable band seating requirements, the lot shall be rejected. If one assembly fails, a second sample consisting of double the number of assemblies in the first sample, shall be tested. If one or more failures occur in the second sample, the lot shall be rejected.

6/ If ferrous metal discontinuity in excess of 0.005 inch across its greatest dimension is found in any assembly of the sample, as witnessed by the test of 4.5.5, the lot will be rejected.

7/ Failure of the contractor to correctly identify all standards used in the test of 4.5.6, will be cause for rejection of all production inspected since the previous inspection effectiveness test.

8/ Failure of the phosphate coating on the standard panels to comply with the applicable requirements, shall be cause for rejection of all production assemblies phosphated since the preceding test.

9/ Failure of two or more assemblies of the sample to comply with the applicable requirements for paint adhesion, shall be cause for rejection of the lot. If one assembly of the sample fails, a second sample consisting of double the number of assemblies in the first sample shall be tested. If one or more failures occur in the second sample, the lot shall be rejected.

10/ Failure of the protective coating on one or more assemblies of the sample to comply with the applicable requirements for corrosion resistance, shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

11/ The lot will be rejected, if any one of the following types of metal parts separation or breakup occurs in the metal parts security test, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s):

a. Body breakup.
b. Adapter breakup or separation.
c. Complete rotating band separation. This is defined as detachment of all or part of the band having band seat knurling imprint. The following acceptance criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint:

(1) The lot will be accepted if 2 or less assemblies throw band fragments having length not greater than 0.500 inch.
(2) The lot will be rejected if 3 or more assemblies throw band fragments having length not greater than 0.500 inch.
(3) The lot will be rejected if any assembly throws a band fragment having length greater than 0.500 inch (indicative of deep-seated band lamination).
(4) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.4.5 Packaging and marking inspection.-Inspection for packing and marking shall be in accordance with MIL-P-10025.

4.4.6 Inspection equipment.-Inspection equipment conforming to the designs of drawings listed on IEL 7259220 shall be used to perform the specified examinations and tests.

4.5 Test methods and procedures.

4.5.1 Hardness.-The methods of test for hardness shall be as specified in 243.1 of Federal Test Method Standard No. 151.

4.5.1.1 Body.-The sample bodies shall be sectioned and tested for hardness as shown on Dwg. D7259222.

4.5.1.2 Adapter.-One hardness impression shall be taken per test assembly on the prescribed adapter area. The bourrelet diameters of the test assemblies shall be supported on a V-notched anvil or double cylinders. No cylindrical correction shall be applied. The test shall not be considered destructive if the maximum acceptable bourrelet diameter at the impression is not exceeded (see 6.4).

4.5.1.3 Nose.-Three hardness impressions shall be taken, approximately 120° apart, on the large outside diameter, base end, per test nose. The average of the three readings shall be considered to be the hardness of a test nose. Test noses shall be supported on a mandrel approximately 0.5 inch in diameter. No cylindrical correction shall be applied. The test shall not be considered destructive if the maximum acceptable cylindrical diameter is not exceeded (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.5.2 Adapter security.-The test for adapter security shall be conducted using a standard tension testing machine of sufficient capacity. Rate of pull shall be 1½ inches ± ¼ inch per minute.

4.5.3 Band tightness.-The band tightness tester (RB-20) shall have an initial ram pressure of 200 pounds per square inch (p.s.i.), and a final ram pressure of 3,000 p.s.i. The average radial void applicable to band tightness shall be one-fourth of the total reading, registered above zero (0), on the tester indicator at final ram pressure.

4.5.4 Soundness.-The inspection process shall be magnetic particle as prescribed in MIL-I-6868 for fluorescent suspension, utilizing the wet, continuous, circular method and a magnetizing current of 1,000 ± 200 amperes, D.C. Electrodes shall make contact with the base end and the mouth or bottom surface of each assembly cavity. The entire exterior ferrous metal surface of each assembly of the lot shall be examined for magnetic particle indications of discontinuity.

4.5.5 Soundness verification.-The inspection sample will be processed as prescribed in 4.5.4. Each assembly of the sample showing a magnetic particle indication of ferrous metal discontinuity, will be sectioned 90° to its longitudinal axis at the approximate midpoint of maximum magnetic particle concentration. An assembly which shows two or more magnetic particle indications will be sectioned in a similar manner, at the single point of maximum magnetic particle concentration. The cross section specimen will be polished and examined at 100 ± 25 diameters for ferrous metal discontinuity. A sample assembly which shows ferrous metal discontinuity that measures more than 0.005 inch across its greatest dimension on the plane of the cross section, will be classed defective. Soundness verification inspection will be conducted by Government personnel.

4.5.6 Magnetic particle inspection effectiveness.-Government personnel will establish, identify, and maintain inspection effectiveness standards to be used as per 4.4.4.1, Table I. A minimum of fifty (50) standards free of any magnetic particle inspection indications, and fifty (50) standards having magnetic particle inspection indications of various magnitude and external location will be provided. All standards, prior to use, will be approved and furnished upon request by the Industrial Engineering Division, Lake City Army Ammunition Plant. Identification marking will be such as to not be visible to the contractor's magnetic particle inspector(s) when the standards are being inspected at the same rate as production inspection. The test sample shall include not less than ten (10) standards with indications, and not less than ten (10) standards free of indications. The test will be conducted under close surveillance of the Government inspector and with no routine fixed time pattern for test frequency. The Government inspector will make every reasonable effort to conduct the test so that the contractor's magnetic particle inspector(s) is/are unaware that the test is under way.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.5.7 Magnetic particle inspection equipment effectiveness.-Government personnel will employ, identify, and maintain a group of five (5) equipment effectiveness standards. All standards, prior to use, will be approved and furnished upon request, by the Industrial Engineering Division, Lake City Army Ammunition Plant. The group will contain, in its entirety, indications of relatively weak magnitude and so dispersed that all exterior portions of the projectile body are represented, at least once, by an indication. The group of standards will be fed into the magnetic particle inspection machine at the start of each shift, and after the starting of the machine following a shutdown exceeding 15 minutes, or at any other time deemed necessary by the Government inspector. Each standard will be allowed to go through the machine, untouched, and be carefully lifted before it is dropped from its support. Failure to observe a known indication, or observance of a mechanical or other interference with any known indication, will be evidence that the machine is not qualified to be used for production acceptance. Standards shall be free of magnetic particles prior to use, as determined by examination under suitably filtered black light.

4.5.8 Phosphate coating.-The method of test shall be as prescribed in TT-C-490 for type coating used in accordance with applicable drawings (see Dwg. D7259220). The prescribed panels shall be processed with assemblies represented at the start of daily production and at four-hour intervals, maximum, thereafter. The coating appearance and weight shall be recorded.

4.5.9 Paint adhesion.-The method of test shall be as prescribed in TT-C-490. The test shall be made on the exterior surface of each assembly of the sample, forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal on any assembly of the sample as a result of the test, shall be recorded as a defect.

4.5.10 Corrosion resistance.-The method of test shall be as prescribed in Method 811.1 of Federal Test Methods Standard No. 151, Metals; Test Methods. The rotating band and base end of the sample assemblies may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.6.3. The presence of ferrous metal corrosion on the exterior surface forward of the rotating band of any assembly of the sample as a result of the test, shall be interpreted and recorded as protective coating failure.

4.5.11 Metal parts security.-The method of test will be as prescribed in AMCR 715-505, Volume 8, as applicable. The sample assemblies will be loaded and assembled as specified on Dwg. D7259192, Cartridge, 20mm, for projectile metal parts test. The test cartridges, at ambient temperature, will be fired in bursts of fifty shots in an M39 gun. The gun barrel will be at ambient temperature for the firing of each burst. The cyclic rate of fire will be recorded for informational purposes. This test will be performed by Government personnel at the testing facility specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.-Painted body-adapter and rotating band assemblies shall be packed in accordance with Dwg. F7258861 or D7259136. Noses, metering discs, and closure discs shall be packed in a manner acceptable to the procuring activity.

5.2 Marking and labeling.-Marking on packs shall be in accordance with specification MIL-P-10025.

5.3 Data cards.-Ammunition data cards shall be furnished. The requirements of MIL-STD-1167 shall apply.

6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements (see 3.2 and 4.3).

6.1.4 Packing instructions (see 5.1).

6.1.5 Instructions for shipment of production lot samples for metal parts security test.

6.1.6 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment, where applicable, for acceptance test purposes.

6.2 First article sample.-The procurement agency may waive the requirement for a first article sample if the contractor has recently demonstrated his ability to produce this item. The contracting officer will notify the testing facility when to proceed with the metal parts security test.

6.3 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.4 Dimensional effect of hardness testing.-The Rockwell A hardness impression on the adapter (body, adapter and rotating band assembly) will increase bourrelet diameter approximately .005 inch at the impression. The Rockwell hardness impression on the nose will increase the base cylindrical diameter less than .0005 inch at the impression.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

6.5 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitations, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.6 Intermediate point inspection.-The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the item.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1305-A428

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.